HB++: Topology-Conditioned Propagation Bases for Shallow Graph Learning

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Abstract

Precomputed propagation lets a shallow graph learner use multi-hop context without increasing message-passing depth, but it leaves a central design choice unresolved: should the same propagated basis be used at every node? We study this question with an auditable adaptive propagation basis. The base model, APB-GeomFull-FiLM, uses a compact core-derived descriptor and four local propagation-geometry statistics to condition bounded, groupwise coefficients around the canonical $[X, PX, P^2X]$ basis. We then introduce APB-GeomTopo-FiLM as a topology-flow extension that adds four node-local degree-flow statistics to the same controller.

The contribution is a controlled basis adaptation mechanism and an evaluation protocol that separates propagation range, capacity, coordinate choice, and structural alignment. The primary evidence is a frozen confirmatory matrix covering 11 public graph datasets and 83 public splits with three seeds per split. Relative to the matched Low+Role reference, APB-GeomFull-FiLM has a positive mean Accuracy increment on 7 of 11 datasets and the largest increment is +1.29 percentage points on Roman-empire; its mean NLL change across dataset-level contrasts is −0.00491. The topology-flow extension is positive on 7 of 11 datasets and reaches +1.39 points on Roman-empire. A leave-one-out audit and degree-matched shuffle show that this Roman increment depends on a joint, node-aligned topology-flow signal rather than on a degree marginal alone. The resulting claim is deliberately scoped: propagation-geometry conditioning provides a broadly transferable shallow basis interface, while topology-flow conditioning is a graph-regime-dependent extension rather than a universally stronger graph learner.

CCS Concepts

• Information systems → Web mining; **• Computing methodologies → Machine learning algorithms**; **• Networks →** *Network algorithms.*

Keywords

node representation learning, graph neural networks, propagation bases, structural roles, precomputation, heterophilous graphs

ACM Reference Format:

Anonymous Author(s). 2027. HB++: Topology-Conditioned Propagation Bases for Shallow Graph Learning. In *ACM International Conference on Web Search and Data Mining (WSDM '27), 2027*. ACM, New York, NY, USA, 10 pages.

1 Introduction

Large graph-learning systems often need multi-hop context but cannot afford to couple every increase in information range to a deeper message-passing network. Precomputed propagation offers a clean separation: graph transforms are materialized once, and a shallow learner consumes the resulting channels [4, 5, 21, 22]. This separation is useful for scalable training, repeated model fitting, and controlled studies of what the learner receives.

The separation also exposes a less studied design question. A fixed basis gives every node the same combination of propagated channels, even when local neighborhood structure varies. Appending a structural vector does not resolve this question: the learner may use the vector as an ordinary input while leaving the propagation basis unchanged. Existing multi-hop learners and polynomial filters mainly learn a global combination of graph transforms or expose a fixed channel bank [2, 4, 8, 15]. Structural encoders provide a second source of graph information, but their usual interface is another representation rather than a control signal for the propagation coordinates [3, 18, 20].

We frame the problem as topology-conditioned basis adaptation for a shallow learner. The base model, APB-GeomFull-FiLM, starts from the canonical channels $[X, PX, P^2X]$ and uses a compact core-derived descriptor together with four label-free propagation-geometry statistics to produce bounded, groupwise coefficients. Its controller is initialized at the fixed basis, so the adaptive model has an explicit reference point under the same backbone, propagation range, and parameter budget. The downstream encoder remains a two-layer GraphSAGE model.

We also study a specific extension rather than treating every structural descriptor as interchangeable. APB-GeomTopo-FiLM adds four local topology-flow statistics: self log-degree, neighbor mean log-degree, neighbor log-degree dispersion, and neighbor inverse-degree mass. The extension asks whether the propagation-geometry controller can benefit from node-aligned local topology flow. The experiments show that this is a selective property. On Roman-empire, the extension produces a reproducible increment relative to the base controller; degree-matched shuffling preserves the degree marginal but removes that increment, and leave-one-out results indicate that the effect is jointly encoded. Other real graphs and the controlled synthetic sweep do not support a simple global descriptor rule.

This paper therefore makes three focused contributions. First, it formulates topology-conditioned propagation basis adaptation as a separate interface for shallow graph learning and gives a bounded, identity-initialized construction. Second, it separates the base propagation-geometry method from the optional topology-flow extension, allowing the extension's operating boundary to be measured instead of folded into a universal performance claim. Third, it provides a fixed-budget audit with matched capacity, propagation range, coordinate, degree, rewiring, and frozen-holdout controls. Its confirmatory evidence is a post-freeze matrix of 11 public graph datasets and 83 public splits, with the dataset list, public masks, model list, seeds, and checkpoint rule frozen before test evaluation. The paper's cross-dataset claims are made only from this matrix, whose dataset list, public masks, model list, seeds, and checkpoint rule form the main evidence contract.

The rest of the paper first positions explicit bases and structural controls, then defines the canonical basis, the base controller, and the topology-flow extension. The experiments establish the value of explicit propagated inputs, separate basis adaptation from capacity, and finally use transfer and intervention evidence to characterize when topology-flow conditioning is useful.

2 Related Work

2.1 Explicit propagation and graph filtering

Graph convolution and GraphSAGE combine feature transformation with learned neighborhood aggregation [7, 11]. SGC and APPNP separate part of this propagation from the learned predictor [5, 22]. SIGN, SAGN, and FSGNN expose multiple precomputed channels for scalable learning or learn how to combine transformed features [4, 15, 21]. Other graph filters learn global propagation weights or mix low- and high-pass signals, including MixHop, GPR-GNN, ACM-GCN+, and ChebNetII [1, 2, 8, 14].

These methods establish the value of making propagation range or filter shape explicit. Their learned combinations are generally global, or their channel banks are fixed before the predictor. Our base method addresses a different interface: it uses graph descriptors to condition the coefficients of a low-order basis at each node and feature group while retaining a shallow backbone and a fixed propagation range. The comparison is thus about where adaptation enters the computation, not about replacing existing graph filters.

2.2 Structural position and topology-conditioned computation

Structural-role methods describe graph position through recurring patterns or structural similarity. Role discovery, struc2vec, and GraphWave construct representations from graph structure rather than immediate attribute proximity [3, 18, 20]; position-aware GNNs provide a related route for exposing graph position to a learned model [24]. In a separate line, FavardGNN, OptBasisGNN, UniBasis, and UniFilter study learned or universal bases for graph filtering and heterophily [6, 9, 10].

Our distinction is the coupling between these two interfaces. A compact descriptor is not only appended as an input and is not used to define a global filter. It controls a bounded deviation from a canonical propagation basis. The paper first evaluates the base geometry-conditioned controller and then tests topology-flow as an extension. This separation is important because the intervention results show that the extension is useful in a graph-dependent, node-alignment-sensitive regime rather than as a universal replacement for fixed propagation.

2.3 Heterophily, depth, and evaluation controls

Heterophilous graph learning provides a complementary perspective on the choice of propagated coordinates. H2GCN separates ego and neighbor embeddings and combines higher-order neighborhoods, while Geom-GCN constructs geometric neighborhoods to address structural information loss and long-range dependence in disassortative graphs [17, 25]. LINKX shows that large non-homophilous graphs also require simple, scalable controls that do not assume a uniformly useful message-passing signal [13]. These works motivate the graph-regime and capacity controls in our evaluation, but they do not study node-wise coefficients for a precomputed multi-hop basis.

A second line of work changes the effective range or the depth of aggregation. Jumping Knowledge networks select representations from different depths, while analyses of graph convolution connect depth to Laplacian smoothing and the loss of node-discriminative information [12, 16, 23]. DropEdge offers a different remedy by modifying the graph during training to reduce the effects of over-smoothing and over-fitting [19]. HB++ holds the downstream backbone and propagation range fixed, materializes the channels once, and asks a narrower question: can graph descriptors control how a shallow learner combines those channels without changing depth or the graph operator? This is the evaluation boundary that separates our interface from depth, graph-augmentation, and heterophily-specific architectural comparisons.

3 Topology-Conditioned Propagation Bases

We separate the method into a fixed propagation interface, a base controller, and a topology-flow extension. This organization keeps the central operation visible: graph descriptors change how a canonical propagation basis is formed, while the downstream learner remains shallow.

3.1 Problem setup and notation

We consider transductive node classification on a directed graph $G = (V, E_D)$ with node attributes $X \in \mathbb{R}^{n \times d}$. Labels are observed only on the training nodes $T \subset V$. The directed edge list is used to define degree and structural descriptors. For attribute propagation, we form the simple symmetric union

$$A = \text{simple}(E_D \cup E_D^{\top}), \qquad P = D^{-1}A, \quad D_{vv} = \max\left(1, \sum_u A_{vu}\right). \tag{1}$$

An isolated node receives a zero neighbor aggregate. The downstream GraphSAGE encoder uses its own neighbor aggregation; P is used for the materialized input basis.

The canonical low-order basis is

$$B = [B^{(0)} \| B^{(1)} \| B^{(2)}] = [X \| PX \| P^2 X]. \tag{2}$$

It exposes the one- and two-hop attribute signals before the trainable encoder. An equivalent residual coordinate system can be written as $[X, (I - P)X, (I - P^2)X]$ because $PX = X - (I - P)X$ and $P^2 X = X - (I - P^2)X$. We use this relation to distinguish a choice of coordinates from a change in the information span.

All materialized channels are standardized column-wise using training nodes:

$$\widetilde{B}_{ij} = \frac{B_{ij} - \mu_j}{\max(\sigma_j, 10^{-6})}. \tag{3}$$

The training statistics are reused for validation and test nodes.

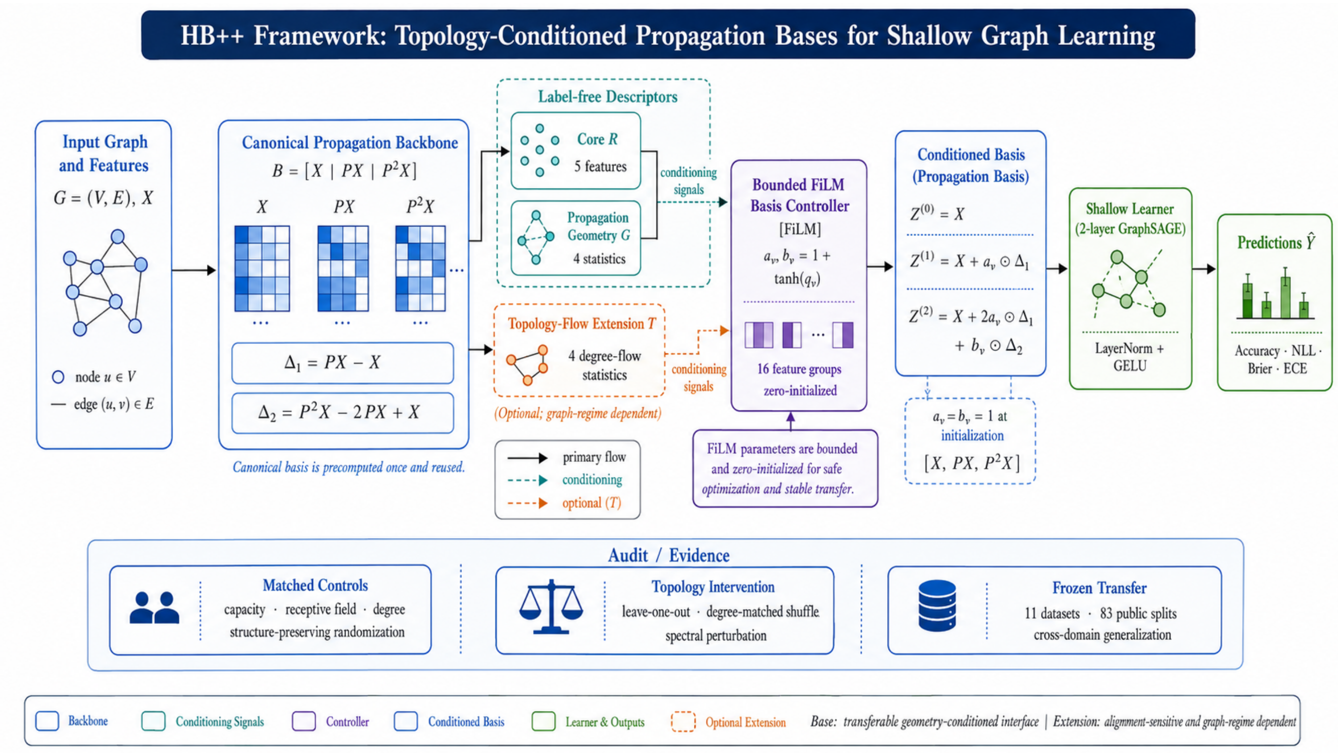

Figure 1: HB++ separates cached propagation coordinates from shallow prediction. Core-derived role and propagation-geometry signals drive the base controller; topology-flow statistics enter through the optional extension. The lower band summarizes the evidence contract used to assess capacity, mechanism, and cross-dataset transfer.

Table 1: Notation used by the adaptive basis.

Symbol	Meaning
P	row-normalized symmetric-union propagation operator
B	canonical basis $[X, PX, P^2X]$
$\mathbf{R}$	five-dimensional core-derived descriptor
$\mathbf{G}$	four propagation-geometry statistics
$\mathbf{T}$	four topology-flow statistics used only by the extension
Δ_1, Δ_2	first- and second-order propagation differences
a_v, b_v	bounded node/group coefficients at node v
$Z^{(0:2)}$	conditioned basis consumed by GraphSAGE

3.2 Core-derived controls

The structural descriptor $\mathbf{R} \in \mathbb{R}^{n\times 5}$ is computed from a multiplicity-preserving symmetric incidence graph. It records a compact core-derived position, its logarithmic scale, a degree-bin-centered deviation, a normalized core position, and a residual degree gap. Degree-bin means, thresholds, and all standardization statistics are fitted on T only. The descriptor uses no node attributes and no labels.

The base controller also receives four label-free propagation-geometry statistics. For node v, they are the log norms of X_v, $(PX)_v$, $(PX-X)_v$, and $(P^2X-2PX+X)_v$, each transformed as $\log(1+\|\cdot\|_2)$. They are computed from the materialized basis and standardized on T; we denote the resulting matrix by $\mathbf{G} \in \mathbb{R}^{n\times 4}$. This gives the base method a local signal about how propagation changes the attribute coordinates without introducing another learned structural encoder.

The extension adds the topology-flow matrix $\mathbf{T} \in \mathbb{R}^{n\times 4}$. Its columns are self log-degree, neighbor mean log-degree, neighbor log-degree standard deviation, and neighbor inverse-degree mass. These values are computed from the graph alone and standardized on T. The extension input is $[\mathbf{G}\|\mathbf{T}]$; the base input is $\mathbf{G}$.

For completeness, the exact feature construction used in the reported runs is shown below. Let c_v be the core number of node v in the multiplicity- preserving simple symmetric-union graph, let d_v be its degree in that graph, and let $b(v)$ be its training-fitted equal-frequency degree bin. If $\bar{c}_b$ is the training-node mean of c_v in bin b, then

$$\mathbf{R}_v = \Big[c_v,\ \log(1+c_v),\ c_v - \bar{c}_{b(v)},\ \frac{c_v}{\max(d_v, 1)},\ \log(1+\max(d_v - c_v, 0))\Big]. \tag{4}$$

The four propagation-geometry coordinates are

$$\begin{aligned} \mathbf{G}_v &= [g_1(v), g_2(v), g_3(v), g_4(v)], \\ g_1(v) &= \log(1 + \|X_v\|_2), \\ g_2(v) &= \log(1 + \|(PX)_v\|_2), \\ g_3(v) &= \log(1 + \|(PX - X)_v\|_2), \\ g_4(v) &= \log(1 + \|(P^2X - 2PX + X)_v\|_2). \end{aligned} \tag{5}$$

The topology-flow vector contains $\log(1 + d_v)$, the mean and standard deviation of $\log(1 + d_u)$ over $u \in N(v)$, and the mean of $1/\max(d_u, 1)$ over the same neighbors. Empty neighborhoods use zero aggregates. All means, bin assignments, and standardization statistics that depend on labels or node subsets are fitted on T only.

The implementation specification is compact: the role tower is $5 \longrightarrow 32 \longrightarrow 32$, the geometry tower is $4 \longrightarrow 32 \longrightarrow 32$ for $APB - GeomFull - FiLM$ and $8 \longrightarrow 32 \longrightarrow 32$ for $APB - GeomTopo - FiLM$, and the FiLM and coefficient heads are $32 \longrightarrow 64$ and $32 \longrightarrow 2m$, respectively, with GELU between hidden layers. The $m = 16$ feature groups use contiguous integer boundaries $\lfloor gd/m \rfloor$ and $\lfloor (g+1)d/m \rfloor$ over the original feature coordinates; coefficients are broadcast within each group. Thus changing from four to eight geometry controls adds $4 \times 32 = 128$ first-layer weights and no new propagation or encoder parameters. The propagation and GraphSAGE neighbor sets both use the same simple symmetric union without self-loops; the two operators are kept separate so the cached basis and the downstream aggregation remain explicit.

3.3 Base controller and topology-flow extension

The base method is APB-GeomFull-FiLM. It uses separate two-layer towers for the structural descriptor and the propagation-geometry controls:

$$h_v^R = \phi_R(\mathbf{R}_v), \qquad h_v^C = \phi_C(\mathbf{C}_v),$$
$$\mathbf{C} = \begin{cases} \mathbf{G}, & \text{for } APB - GeomFull - FiLM, \\ [\mathbf{G} \| \mathbf{T}], & \text{for } APB - GeomTopo - FiLM. \end{cases} \tag{6}$$

The geometry tower produces a scale and shift for the role representation:

$$u_v = h_v^R \odot (1 + \gamma(h_v^C)) + \beta(h_v^C), \qquad q_v = W_u u_v + c_u. \tag{7}$$

The final layer producing q_v is zero-initialized. Consequently the controller begins at the canonical basis and learns only a bounded deviation from it. With $m = 16$ feature groups and span $\rho = 0.75$, the coefficients are

$$[a_v, b_v] = 1 + \rho \tanh(\operatorname{reshape}(q_v, 2, m)). \tag{8}$$

Define the propagation differences

$$\Delta_1 X = PX - X, \qquad \Delta_2 X = P^2 X - 2PX + X. \tag{9}$$

The coefficients are shared within each feature group and broadcast across its coordinates. The conditioned basis at node v is

$$\begin{aligned} Z_v^{(0)} &= X_v, \\ Z_v^{(1)} &= X_v + a_v \odot (\Delta_1 X)_v, \\ Z_v^{(2)} &= X_v + 2a_v \odot (\Delta_1 X)_v + b_v \odot (\Delta_2 X)_v. \end{aligned} \tag{10}$$

When $q_v = 0$, $a_v = b_v = 1$ and Eq. 10 recovers $[X, PX, P^2X]$. This identity initialization provides the matched reference point for both the base method and the topology-flow extension.

APB-GeomTopo-FiLM is therefore an extension of APB-GeomFull-FiLM, not a separate backbone. It keeps the same basis, groups, span, optimizer, and shallow encoder, and adds only the four topology-flow controls to the geometry tower. In the reported implementation this changes the controller by 128 parameters.

Algorithm 1 Audited adaptive-basis evaluation

Require: directed edges E_D, features X, train/validation/test masks T, V, U, variant flag $s \in$ {Full, Topo}

1: $A \leftarrow \operatorname{simple}(E_D \cup E_D^\top)$; $P \leftarrow D^{-1}A$
2: $B \leftarrow [X, PX, P^2X]$; compute $\mathbf{R}, \mathbf{G}$ from G
3: **if** s = Topo **then** compute $\mathbf{T}$ and set $\mathbf{C} \leftarrow [\mathbf{G} \| \mathbf{T}]$
4: **else**set $\mathbf{C} \leftarrow \mathbf{G}$
5: **end if**
6: Fit bins and scalers on T; initialize the coefficient head at zero
7: $q \leftarrow \operatorname{Controller}(\mathbf{R}, \mathbf{C})$; form a, b and Z by Eq. 8–10
8: Train the fixed two-layer GraphSAGE on T; retain the checkpoint with lowest validation NLL
9: Fit temperature on validation logits only; evaluate once on U
10: Write metrics, split/source hashes, checkpoint id, and finite-value audit

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3.4 Shallow encoder and computation

The final input is $[\widetilde{Z}^{(0)} \| \widetilde{Z}^{(1)} \| \widetilde{Z}^{(2)} \| \widetilde{\mathbf{R}}]$, with width $3d + 5$. A two-layer GraphSAGE encoder then applies

$$\begin{aligned} m_v^{(\ell)} &= |N(v)|^{-1} \sum_{u \in N(v)} h_u^{(\ell)}, \\ h_v^{(\ell+1)} &= \operatorname{LN}\Big(\operatorname{GELU}\Big(W_\ell [h_v^{(\ell)} \| m_v^{(\ell)}]\Big)\Big), \\ h_v^{(0)} &= z_v. \end{aligned} \tag{11}$$

Dropout is applied between layers and an MLP head produces class logits. The training objective is cross-entropy on T plus $10^{-4} \| [a_v, b_v] - 1 \|_2^2$; validation NLL selects the checkpoint.

The basis requires $O(K|E|d)$ sparse work and $O(Knd)$ storage for K materialized channels. The core decomposition is computed once and the descriptor is reused across training epochs. These costs are separated from epoch-wise aggregation in the resource audit. The extension changes the descriptor computation and controller input but not the propagation order or the downstream message-passing stack.

Algorithm 1 summarizes the frozen evaluation path. It separates graph-side precomputation from controller fitting and makes the validation-only checkpoint and calibration operations explicit.

4 Experimental Setup

4.1 Evaluation questions

The evaluation follows the method hierarchy defined in Section 3. RQ1 asks whether explicit propagated inputs improve a fixed shallow learner. RQ2 asks whether conditioning the propagation basis adds value beyond a fixed descriptor and a fixed basis at matched capacity. RQ3 asks when the topology-flow extension is useful and whether its increment depends on node-aligned local structure rather than on a degree marginal or a single scalar descriptor.

4.2 Datasets and split contract

The post-freeze confirmatory matrix fixes the dataset list, public masks, model list, seeds, and stopping rule before any test evaluation. It contains Cora, CiteSeer, PubMed, Cornell, Texas, Wisconsin, Chameleon, Squirrel, Actor, Roman-empire, and Amazon-ratings. The first three use the public Planetoid split; the remaining datasets use all ten public masks shipped by the PyG loaders, giving 83 public splits in total. Each split uses three training seeds. Accuracy is the primary task metric, while NLL, Brier, and ECE are probability-quality diagnostics. The complete matrix, source-tree hashes, split hashes, and runner hash are in the separate appendix.

The five-graph heterophily transfer set (Roman-empire, Amazon-ratings, Minesweeper, Tolokers, and Questions) supplies an additional mechanism audit with its own per-split ledgers. It is not merged with the post-freeze matrix, whose frozen protocol is the source of the broad cross-dataset claim.

4.3 Models and controls

All methods use a two-layer GraphSAGE encoder with width 256, LayerNorm, GELU, dropout 0.35, and an MLP head. AdamW uses learning rate 10^{-3}, weight decay 10^{-5}, gradient clipping 1.0, at most 60 epochs, and patience 10. The lowest validation-NLL checkpoint is evaluated once on the test set.

The fixed-input references are Raw, Low, and Low+Role. The two adaptive models are the base controller APB-GeomFull-FiLM and its topology-flow extension APB-GeomTopo-FiLM. They share the propagation basis, encoder, group count, span, optimizer, and checkpoint rule. The extension adds four topology-flow controls and 128 controller parameters. Wider Raw and parameter-matched Raw GraphSAGE controls separate basis utility from capacity; matched-receptive-field runs separate explicit propagation from simply increasing neural depth.

The mechanism audit uses two interventions. Exp143 removes each topology-flow component, or the complete topology-flow block, while keeping controller width and parameter count fixed. Exp144 retains self log-degree and shuffles the three neighbor-derived controls within self-degree bins, preserving the degree marginal while disrupting node-aligned local correspondence. Both analyses use the same data and optimization protocol as the complete extension.

4.4 Statistical and integrity protocol

Training seed is the primary replication unit. We report paired means, sample standard deviations, seed bootstrap intervals, and sign-flip checks for the main contrasts. Node-level utilities are descriptive and retain the node as the unit of diagnostic analysis rather than treating pooled nodes as independent graph replications. Temperature is fitted from validation logits only and never uses test labels. All reported result files record source and split hashes and pass finite-value checks. Full official-core adapters, rewiring manifests, capacity ledgers, and per-seed records are supplied in the separate appendix. The 7/11 positive Accuracy count is a descriptive dataset-level summary; it is not a pooled-node significance test, and the public split/seed cells are not treated as independent graph populations.

5 Results

The results are organized around the frozen cross-dataset matrix and the mechanism controls that explain its pattern. The matrix is the paper's primary evidence for the shallow basis interface; the intervention and transfer analyses then characterize when topology-flow conditioning adds value.

5.1 Post-freeze confirmation across graph families

The confirmatory matrix tests the same frozen interface across graph families. It contains 11 public datasets and 83 public splits, with three paired training seeds and the same shallow GraphSAGE backbone for Raw, Low+Role, APB-GeomFull-FiLM, and APB-GeomTopo-FiLM. Test predictions are evaluated once after validation-NLL checkpoint selection; temperature is fitted from validation logits only. The protocol was frozen before this matrix was run, and every split passed the source, mask, model/seed, and finite-value audits.

Table 2 is the detailed method comparison for this frozen contract. It reports both task performance and probability quality under one shared backbone, making the comparison about the propagated basis and its controller rather than about a mixture of model depths or training protocols. Raw and Low+Role provide the fixed-input references; APB-GeomFull-FiLM and APB-GeomTopo-FiLM are the two adaptive basis interfaces.

Figure 2 makes the probability-quality axis explicit: the sign pattern is graph- and metric-dependent, with negative values favorable, so it is read alongside rather than substituted for the Accuracy comparison. The corresponding numeric deltas are reported in Table 3.

Relative to Low+Role, APB-GeomFull-FiLM improves mean Accuracy on 7 of 11 datasets. The largest increment is on Roman-empire (+1.29 percentage points), with additional positive increments on Cora (+0.60), PubMed (+0.20), Cornell (+0.18), Chameleon (+0.12), Squirrel (+0.08), and Amazon-ratings (+0.03). The mean dataset-level NLL contrast is −0.00491, and NLL is lower on 9 of 11 datasets. This pattern supports a transferable base interface while preserving the graph-dependent character of the task metric.

The topology-flow extension is positive on 7 of 11 datasets and reaches +1.39 Accuracy points on Roman-empire. Its mean dataset-level NLL contrast is −0.00527. The extension therefore adds a useful structural control in some regimes, while the base geometry-conditioned interface carries the broader claim. The per-split ledgers and complete probability-quality matrix remain in the separate appendix.

The capacity closure provides a direct check on the explicit-propagation comparison. In the five-seed fixed-split control, Low uses 405,800 parameters and obtains 71.136 ± 0.064% Accuracy. A wider Raw GraphSAGE uses 559,528 parameters and obtains 70.171 ± 0.053%, while a hidden-width-320 Raw control is approximately parameter matched at 404,840 parameters and obtains 69.827 ± 0.047%. The controls are reported here as a capacity audit, not as another cross-dataset leaderboard: they show that the Low row's input-span comparison is not explained by simply giving Raw a larger hidden layer.

5.2 Topology-flow is a selective extension of the base controller

Within the frozen matrix, APB-GeomTopo-FiLM is evaluated as a compact extension of APB-GeomFull-FiLM, adding 128 controller parameters while preserving the basis, encoder, propagation range, and checkpoint rule. It has a positive Accuracy contrast on 7 of 11 datasets and its clearest increment is on Roman-empire. This is the appropriate role of the topology-flow block: a structural control whose utility is measured relative to a strong geometry-conditioned base, not a replacement for the base method on every graph.

Table 2: Task and NLL comparison on the post-freeze broad confirmatory matrix. Each cell reports mean ± sample standard deviation over the public split/seed cells; Accuracy is in percent and lower NLL is better. All four methods use the same two-layer GraphSAGE backbone, optimizer, validation-NLL checkpoint rule, and validation-only temperature fitting.

Dataset	Splits	Raw		Low+Role		APB-GeomFull-FiLM		APB-GeomTopo-FiLM	
		Acc. (%)	NLL	Acc. (%)	NLL	Acc. (%)	NLL	Acc. (%)	NLL
Cora	1	60.57 ± 0.46	1.200	60.20 ± 0.85	1.248	60.80 ± 0.40	1.240	60.77 ± 0.06	1.241
CiteSeer	1	42.70 ± 1.54	1.517	47.07 ± 1.53	1.447	46.73 ± 1.30	1.445	46.70 ± 1.40	1.444
PubMed	1	52.57 ± 0.87	0.992	55.10 ± 1.22	0.968	55.30 ± 1.90	0.965	55.33 ± 1.68	0.965
Cornell	10	45.23 ± 7.17	1.379	43.51 ± 5.87	1.416	43.69 ± 5.68	1.419	43.69 ± 5.68	1.419
Texas	10	65.32 ± 8.67	0.971	65.05 ± 8.60	0.995	64.77 ± 8.41	0.995	64.59 ± 8.41	0.994
Wisconsin	10	59.22 ± 6.56	1.170	56.34 ± 5.94	1.190	55.95 ± 6.00	1.186	55.88 ± 6.07	1.186
Chameleon	10	57.92 ± 2.78	1.089	56.71 ± 2.78	1.134	56.83 ± 2.75	1.133	56.83 ± 2.73	1.132
Squirrel	10	45.12 ± 1.55	1.334	48.17 ± 2.40	1.342	48.25 ± 2.27	1.339	48.24 ± 2.27	1.339
Actor	10	33.24 ± 1.08	1.492	29.26 ± 1.34	1.578	29.22 ± 1.35	1.578	29.23 ± 1.35	1.578
Roman-empire	10	77.81 ± 0.55	0.698	76.73 ± 0.47	0.735	78.01 ± 0.48	0.699	78.12 ± 0.49	0.696
Amazon-ratings	10	47.29 ± 0.60	1.285	47.81 ± 0.70	1.258	47.84 ± 0.72	1.258	47.85 ± 0.73	1.258

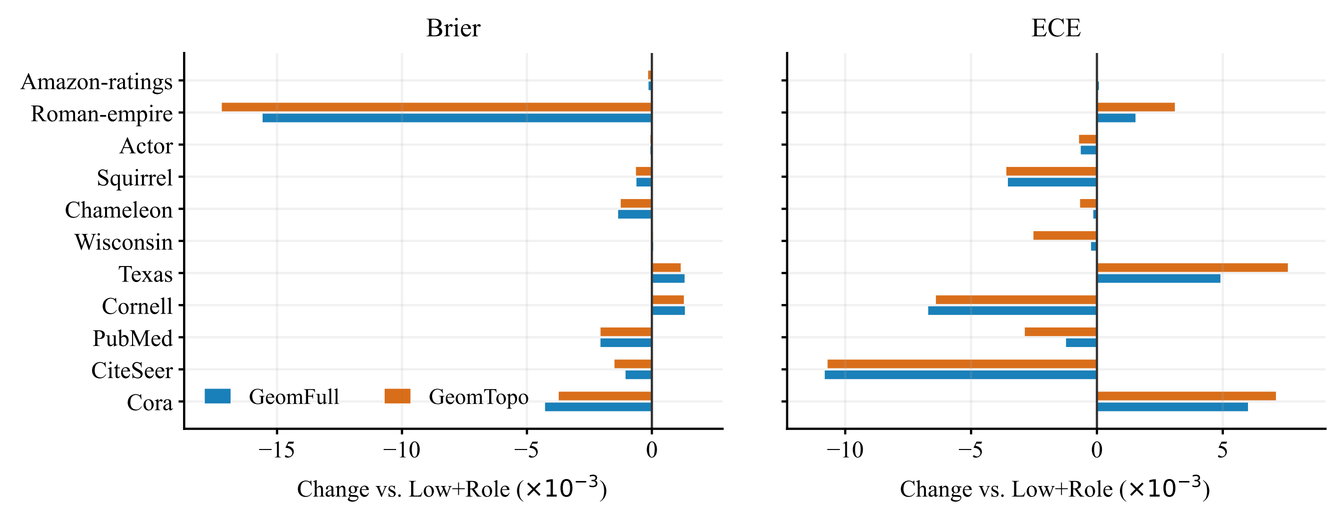

Figure 2: Probability-quality contrasts on the frozen broad matrix. Bars show dataset-level changes relative to Low+Role in Brier score and expected calibration error; negative values are favorable. The plot uses the same audited aggregates as Table 3 and is intended to separate probability quality from the primary task metric.

5.3 Transfer locates the topology-flow increment

The five-graph transfer matrix gives an additional mechanism boundary map. Relative to APB-GeomFull-FiLM, APB-GeomTopo-FiLM improves Roman-empire Accuracy by 0.177 points, NLL by 0.002737, Brier by 0.001378, and ECE by 0.000334 in the paired calibrated comparison. The corresponding changes on Amazon-ratings, Minesweeper, Tolokers, and Questions are close to zero in the probability metrics and primary task metric. This transfer closure uses the separate five-graph ledger with ten fixed public masks and three paired training seeds per mask; it is not the same protocol as the 83-cell broad matrix above. The five graphs therefore support a scoped statement: topology-flow can add value in a particular structural regime, while global graph statistics alone do not identify that regime.

The equal-capacity CiteSeer/PubMed intervention supplies a complementary boundary case: removing topology flow changes NLL by only −0.000032 and −0.000002, respectively. These results show that the controller interface can be retained without requiring every graph to expose a measurable topology-flow increment. Validation-only temperature scaling is reported as a separate calibration operation and does not change Accuracy.

Table 3: Probability-quality deltas relative to Low+Role on the post-freeze broad confirmatory matrix. Negative Brier and ECE changes are favorable; values are dataset-level means over the public split/seed cells.

Dataset	Splits	Δ Brier Full	Δ Brier Topo	Δ ECE Full	Δ ECE Topo
Cora	1	-0.0043	-0.0037	+0.0060	+0.0071
CiteSeer	1	-0.0011	-0.0015	-0.0108	-0.0107
PubMed	1	-0.0021	-0.0021	-0.0012	-0.0029
Cornell	10	+0.0013	+0.0013	-0.0067	-0.0064
Texas	10	+0.0013	+0.0012	+0.0049	+0.0076
Wisconsin	10	+0.0001	+0.0001	-0.0003	-0.0025
Chameleon	10	-0.0014	-0.0013	-0.0002	-0.0007
Squirrel	10	-0.0006	-0.0007	-0.0036	-0.0036
Actor	10	-0.0001	-0.0001	-0.0007	-0.0007
Roman-empire	10	-0.0156	-0.0172	+0.0016	+0.0031
Amazon-ratings	10	-0.0002	-0.0002	+0.0001	-0.0000

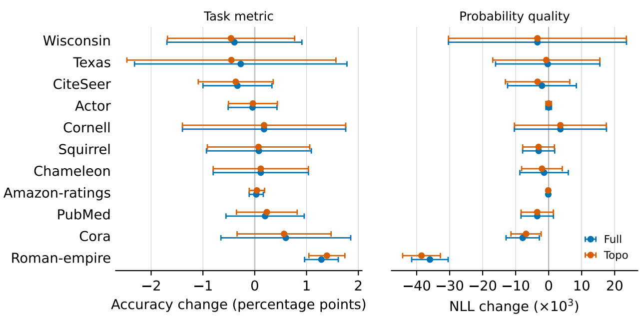

Figure 3: Post-freeze broad confirmatory matrix. Points show candidate minus Low+Role; horizontal bars show sample variation across public split/seed cells. The base controller has a positive mean Accuracy contrast on 7 of 11 datasets, while NLL improvements are more widespread.

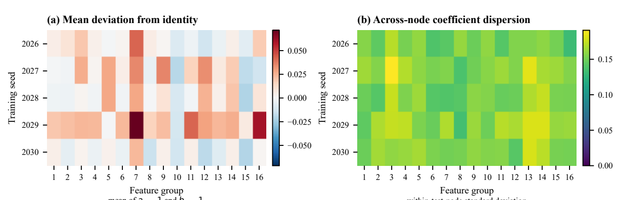

Figure 4: The adaptive controller learns groupwise deviations from the canonical basis. The plot is descriptive evidence that the controller is active; the intervention results below provide the structural mechanism test.

The paired transfer values are collected in Table 4.

Figure 5 places the two main controls on the same scale: the capacity panel isolates the propagated input span from hidden width, whereas the transfer panel locates the topology-flow increment across the five graphs with a paired protocol.

5.4 Official model cores

Pinned author-repository cores are included as adapter-controlled references, not as a native leaderboard. The main text retains

Table 4: Topology-flow extension relative to the base controller. Deltas are APB-GeomTopo-FiLM minus APB-GeomFull-FiLM; negative NLL/Brier is favorable. The Roman row is the clearest transfer effect.

Dataset	Primary delta	NLL	Brier	ECE
Roman-empire	+0.177 pp	-.002737	-.001378	-.000334
Amazon-ratings	-.003 pp	-.000002	-.000003	+.000110
Minesweeper	-.004 pp	+.000081	+.000088	+.000074
Tolokers	+.012 pp	-.000051	-.000087	-.000555
Questions	+0.0003 pp	+.000027	+.000014	+.000045

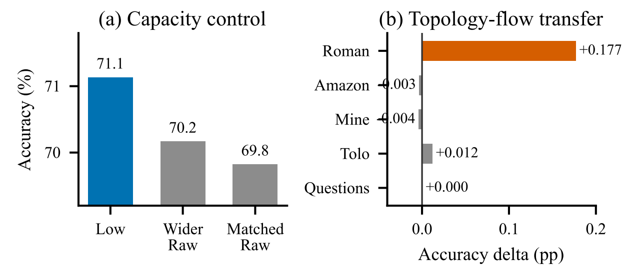

Figure 5: Capacity and transfer controls for the adaptive-basis comparison. The left panel separates the propagated input span from hidden width; the right panel shows that topology-flow is selective, with its clearest increment on Roman-empire. Values are copied from the audited ledgers.

the focused shallow comparison; the complete core matrix, ChebNetII/PPRGo sanity checks, and repository commit manifest are in the appendix. This keeps the central result about basis conditioning rather than a heterogeneous collection of adapter choices.

6 Mechanism and Operating Boundary

The transfer matrix identifies where the topology-flow extension changes the prediction. The intervention audit asks what property of the topology input is used in that case. This section is organized around three observations: the extension is not universally useful, its clearest effect depends on node alignment, and the effect is jointly encoded across the topology-flow block.

6.1 Topology-flow is not explained by a global descriptor

Exp141 pairs graph-level descriptors with the APB-GeomTopo-FiLM–APB-GeomFull-FiLM contrast. Across five real graphs, no single global statistic gives a stable rule. For example, Amazon-ratings and Questions have larger degree heterogeneity than Roman-empire, yet their topology-flow increments are close to zero. Tolokers has strong local topology variation without a comparable Accuracy change. The descriptive sample contains only five graph-level observations, so these statistics are used to locate the operating boundary, not to fit a selector or to claim an inferential graph-regime law.

Exp145 reaches the same conclusion under controlled generation. Varying homophily, degree heterogeneity, endpoint rank coupling, and feature SNR over 16 regimes and three graph seeds produces an all-cell raw mean NLL difference of +0.000071 for APB-GeomTopo-FiLM minus APB-GeomFull-FiLM. One validation-temperature boundary event dominates the unfiltered calibrated mean; four fresh graph seeds for that regime do not reproduce it, and the 47-cell sensitivity mean is −0.000010. The synthetic family is therefore useful as a guard against a simple phase-law claim, not as evidence for a universal predictor of utility.

6.2 When the increment appears, node alignment matters

Roman-empire provides the clearest intervention sequence. The complete topology-flow block improves over the base controller by +0.177 Accuracy points, −0.002737 NLL, and −0.001378 Brier. Exp144 retains the self log-degree marginal but shuffles neighbor-derived topology-flow values within self-degree bins. This degree-matched shuffle changes the Roman result by −0.271 Accuracy points, +0.001391 NLL, and +0.000830 Brier relative to the original aligned controls. The comparison preserves a degree marginal while disrupting node-local correspondence, so the observed effect is tied to the alignment of local topology flow with the node rather than to degree alone.

This is an intervention-based sensitivity result, not a causal identification of one graph property. Its value is narrower and more direct: it distinguishes the node-aligned topology-flow interface from a degree-marginal shortcut under the same controller and split.

6.3 The topology-flow block acts jointly

Exp143 removes one topology-flow component at a time while preserving the controller width and parameter count. On Roman-empire, removing all four components changes Accuracy by −0.241 points and NLL by +0.002734 relative to the complete block. Removing neighbor mean log-degree gives the largest single-component NLL change, +0.000593, while removing the other individual components produces smaller changes. No single scalar reproduces the complete block contrast. The evidence is consistent with a joint function

$$U_v \approx f\left(\log d_v, \mathbb{E}_{u\in N(v)}[\log d_u], \mathrm{Std}_{u\in N(v)}[\log d_u], \mathbb{E}_{u\in N(v)}[d_u^{-1}]\right), \tag{12}$$

where U_v denotes conditional utility and the expression is a mechanism description rather than a fitted utility model.

On CiteSeer and PubMed, the same equal-capacity leave-one-out and shuffle interventions remain near zero. These graphs serve as boundary cases in the argument: the controller can retain the topology-flow interface without requiring that every graph expose a measurable topology-conditioned increment. The node-level Exp142 analysis likewise shows heterogeneous per-node utility across seeds and descriptors, but it does not support a stable monotone rule.

Together, the three analyses support a scoped mechanism statement. Topology-flow conditioning is an alignment-sensitive, graph-regime-dependent augmentation of propagation geometry. The

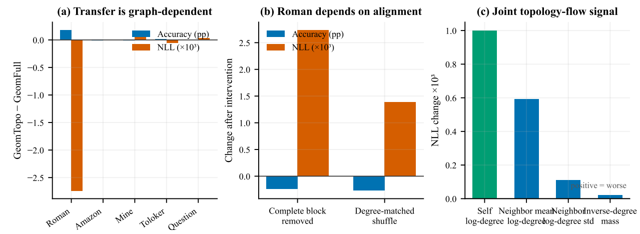

Figure 6: Mechanism audit for the topology-flow extension. Panel (a) shows that the transfer increment is concentrated on Roman-empire. Panel (b) shows the Roman changes after complete removal and degree-matched shuffling. Panel (c) shows that no single topology-flow component reproduces the full block effect. Values are computed from the audited Exp141, Exp143, and Exp144 artifacts.

current evidence does not support a learned graph selector or a universal regime law, so neither is part of the method.

7 Discussion and Scope

The paper's main design choice is an interface: precompute a small propagation basis, then let graph descriptors condition how that basis is combined by a shallow learner. APB-GeomFull-FiLM is the base method for this interface. APB-GeomTopo-FiLM extends it with local topology flow when that signal is available to the controller. The two-level formulation keeps the method usable even when the extension's structural increment is small.

The practical workflow is simple. The graph operator, propagated channels, core-derived descriptor, and topology-flow controls are computed once. The shallow encoder is then trained with ordinary checkpoint selection, while the controller configuration, split hashes, and selected checkpoint are retained with the result. The extension changes the controller input, not the message- passing stack or the propagation order.

The two main tables should be read as an audited matrix rather than as a pool of independent graph observations. It contains 83 public split cells, three training seeds per cell, and four shared-backbone methods, yielding 996 model–split–seed records. Cora, CiteSeer, and PubMed contribute one public Planetoid split each; the other eight datasets contribute ten public masks each. Aggregation is performed within dataset before the cross-dataset count is reported, so the 7/11 positive Accuracy summary is descriptive and does not treat nodes, seeds, or masks as independent graph populations. The machine-readable appendix records the per-split ledgers together with source, split, runner, and finite-value checks. This reporting contract is what lets the broad table test portability of the interface without turning repeated masks into extra datasets.

The same separation applies to the resource profile. The K propagated channels require one sparse pass with $O(K|E|d)$ work and $O(Knd)$ materialized storage, after which the graph-side state is reused across epochs. The topology-flow extension adds four controller inputs and 128 parameters; it does not add propagation steps, change the receptive field, or introduce a deeper message-passing

Table 5: Operational and audit profile of the frozen confirmatory matrix.

Item	Frozen contract or retained artifact
Coverage	11 datasets, 83 public split cells, 3 seeds, and 4 shared-backbone methods (996 records)
Split policy	One public Planetoid split for Cora/CiteSeer/PubMed; ten public masks for each of the other eight datasets
Learner control	Same two-layer GraphSAGE backbone, optimizer, validation-NLL checkpoint rule, and validation-only temperature operation
Graph-side state	$[X, PX, P^2X]$ and descriptors computed once; $O(K\|E\|d)$ sparse work and $O(Knd)$ materialized storage
Extension cost	Four topology-flow inputs and 128 controller parameters; no added propagation step or message-passing layer
Integrity	Source, split, runner, and finite-value checks retained with per-split ledgers

stack. Thus the detailed comparison isolates the conditioning interface while the appendix carries the per-seed capacity and precomputation ledgers. Validation-only temperature fitting is a separate post-training operation for NLL, Brier, and ECE and never changes Accuracy or uses test labels.

Table 5 summarizes the resulting frozen contract and the retained integrity artifacts.

7.1 What the comparison isolates

The shared-backbone design gives the matrix a more specific interpretation than a conventional collection of architecture scores. Raw and Low+Role differ in the propagated coordinates supplied to the learner, while APB-GeomFull-FiLM and APB-GeomTopo-FiLM add a controller whose input is fixed before training. The encoder, optimizer, checkpoint rule, and split policy are held constant across these rows. A contrast in this table can therefore be traced to the representation interface and its controller, with model capacity and graph depth kept out of the primary comparison. The appendix records the wider baselines and released cores for readers who want a different comparison boundary.

This design also clarifies how the three evaluation axes should be read. Accuracy measures the task decision produced by the shallow learner. NLL and Brier measure the quality of its probability output, and ECE summarizes one calibration partition of that output. They are related but not interchangeable: an interface can preserve the task decision while changing confidence, or it can improve the task metric without making every reliability statistic move in the same direction. Reporting the axes together makes the contribution testable under both the prediction and probability interfaces without assigning one metric responsibility for the others.

Finally, the matrix is designed for reuse rather than only for a single reported mean. Each cell retains the source-tree hash, public split or mask hash, model and seed ledger, checkpoint identifier, and finite-value checks. This lets a future reader recompute a dataset-level contrast, inspect whether a result is driven by one public mask, and separate the fixed graph-side cost from the per-epoch learner cost. The resulting audit trail is part of the experimental interface: it connects the basis construction, the controller comparison, and the mechanism interventions to the same frozen evidence contract.

The evidence defines the intended scope. The strongest mechanism evidence comes from Roman-empire, where a degree-matched intervention separates node-aligned local topology flow from a degree marginal. The same interface is not assigned a universal graph-level benefit: the frozen 11-dataset matrix shows positive task-metric contrasts on 7 datasets for each controller, while probability-quality gains are more widespread. CiteSeer, Texas, Wisconsin, and Actor provide useful boundary cases, and the controlled synthetic sweep keeps the mechanism claim selective. This is useful for deployment because it turns a structural descriptor into an auditable optional control signal rather than a hidden assumption about every graph.

The matrix also separates two kinds of evidence that are often conflated in shallow graph comparisons. Accuracy is the task-facing outcome, whereas NLL, Brier, and ECE describe the quality of the probability interface. The base controller improves Accuracy on 7 of 11 datasets and has a lower mean NLL contrast on 9 of 11; the topology-flow extension has the same positive-count pattern but a more localized task increment. This supports a layered reading of the method: geometry-conditioned basis selection is the reusable interface, and topology flow is an optional refinement whose value is concentrated in a particular structural regime. The probability metrics are reported to expose that distinction, not to convert every small calibration change into a claim of universal calibration improvement.

This interpretation also determines how the appendix is used. The appendix expands the main comparison with per-seed ledgers, capacity controls, official core adapters, rewiring manifests, and exact hashes. Those artifacts make the reported contrasts inspectable, while the main paper keeps the visual center on the two shared-backbone comparisons and the paired topology-flow interventions. In particular, the additional controls are evidence about alternative explanations for the interface, not extra leaderboard entries.

The study targets transductive node classification with fixed first- and second-order propagation. The cross-dataset claims are tied to the frozen matrix and its public split contract; the interventions provide matched-protocol sensitivity evidence rather than causal identification.

8 Conclusion

We study how a shallow graph learner should use precomputed multi-hop information when local graph structure varies across nodes. The base method, APB-GeomFull-FiLM, conditions a bounded propagation basis on compact structural and propagation-geometry descriptors while retaining the canonical $[X, PX, P^2X]$ reference and an ordinary GraphSAGE encoder.

The topology-flow extension, APB-GeomTopo-FiLM, adds a second structural interface with a clearly defined operating boundary. Its clearest gain appears on Roman-empire. Leave-one-out and degree-matched shuffling show that this gain is associated with a joint, node-aligned local topology signal rather than with a degree marginal or one scalar feature. The remaining transfer and synthetic results keep the claim selective rather than universal.

The central message is that propagation bases can be conditioned without deepening the learner. Across 11 datasets and 83 public splits, the matrix supports a transferable shallow basis interface; topology-flow remains an auditable extension for particular regimes, clearest on Roman-empire.

9 Ethical Considerations

The study uses public benchmark graphs and does not collect personal data or make decisions about individuals. Nevertheless, node representations can be deployed in social or information systems where graph structure reflects historical bias, missing links, or unequal observation. Structural-role features may amplify such patterns even when node attributes are removed. Applications should therefore evaluate subgroup error, provenance, and the consequences of topology-based inference in their own context. Precomputation also shifts rather than eliminates computational cost. Reporting wall-clock time, memory, hardware, and carbon-relevant resource use is necessary before claiming an efficiency advantage. We report negative controls and rejected hypotheses to reduce selective presentation of favorable results.

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